

SATENDRA YADAV

Aerospace Engineer

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Aerospace Engineering graduate and CFD Trainee Engineer at HAL, specializing in ANSYS Fluent and CFD++ for aerodynamic analysis. Experienced in CFD modeling, SpaceClaim/MIME meshing, CATIA V5, and GD&T, with hands-on exposure to aerospace R&D and engineering validation.

SKILLS

CATIA V5 | CREO | SpaceClaim | CFD++ | MIME (meshing) | ANSYS Fluent & FEA | Python | MATLAB | C/C++ | GD&T (ASME Y14.5) | CMM Inspection | MS Excel

EMPLOYMENT HISTORY

Trainee Engineer (Apprentice) | Hindustan Aeronautics Limited (HAL)
Aircraft Division, Nashik

Feb 2026 - Present

Research & Development – Aircraft Upgradation

- Performed CFD analyses for missile aerodynamics using ANSYS Fluent and CFD++, covering Captive, Store Separation, Plume, Gust, and Thermal cases.
- Prepared simulation models and generated high-quality computational meshes using ANSYS SpaceClaim and MIME and evaluated pressure, velocity, temperature and flow-field distributions to support aerodynamic and thermal performance assessment.

Summer Intern | Bharat Heavy Electricals Limited (BHEL), Jhansi

Jun 2023 - Jul 2023

- Supported turbine blade design activities and developed knowledge of fluid flow, thermal, and turbomachinery concepts relevant to aerospace CFD.

PROJECTS

Automated Fault Detection System | 1st Place – Capstone

Mar 2025

- Developed an IoT-based predictive maintenance system with real-time monitoring, data-driven fault detection, and automated alerts to improve fleet reliability and reduce downtime.

Aerodynamics Analysis of Aircraft Wings (NACA 2412 Airfoil)

- Performed ANSYS CFD and wind-tunnel validation to evaluate lift, drag, and stall characteristics, achieving a 12% improvement in L/D ratio and 18% reduction in drag coefficient through design iterations.

Pulse Detonation Engine (PDE) Design — Published Patent

Jan 2025

- Published Patent** | Patent Application No. 202411103490A — Designed and patented a PDE propulsion system, applying fluid dynamics and thermodynamics to optimize combustion efficiency, thrust-to-weight ratio, and detonation stability.

COURSES

100 Days of Code: The Complete Python Pro Bootcamp

Apr 2026 - Present

Flight Mechanics & MATLAB/Simulink for Aerospace Engineering at Udemy

Jul 2026 - Present

EDUCATION

Bachelor Of Technology | Lovely Professional University, Jalandhar

2021 - 2025

Aerospace Engineering with a Minor in Aircraft Maintenance